Effective height bounds for singular moduli with a supersingular proximity term

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Allowed standard prerequisites (editorial).

- p-adic Hodge theory of CM elliptic curves, G-functions and height principles. Original complete bibliography and all source-local relation constructions retained.

1 Introduction

In this installment of our series, we study the case of a 1-parameter family of elliptic curves and the values of the associated G-functions at points whose fibers are CM elliptic curves.

1.1 Main Result

The main geometric picture we consider is that of a family $f : \mathcal{E} \to S$ of elliptic curves over a smooth irreducible curve S defined over a number field K. We consider also a point $s_0 \in S(K)$ together with a uniformizer $x \in K(S)$ at s_0. By work of Y. André, see [And89], we may associate to the above picture a family of G-functions "centered at s_0".

Given a place v of a number field L we denote by $\iota_v : L \hookrightarrow \mathbb{C}_v$ the embedding associated to it. Towards the above direction, our main result, modulo some technical considerations, may be summarized as the following:

Theorem 1.1. *Let $f : \mathcal{E} \to S$ and s_0 be as above and assume that $\mathcal{E}_{s_0}$ is CM. Let Y_G be the family of G-functions associated to the pair $(\mathcal{E} \to S, s_0)$.*

If $v \in \Sigma_{K(s)}$, a place of $K(s)$, is such that s is v-adically close to s_0, then there exists a polynomial $R_{s,v} \in \bar{\mathbb{Q}}[\underline{X}]$ for which $\iota_v(R_{s,v}(Y_G(s))) = 0$ and $R_{s,v}(Y_G(x)) \neq 0$ on the functional level.

Moreover, if v is a place of good ordinary reduction of the fiber $\mathcal{E}_{s_0}$ then $R_{s,v}$ is independent of such v.

Remark 1.2. *By "v-adically close" we mean that s is within the radius of convergence of the power series in the family Y_G of G-functions associated to our pair.*

The problem of constructing polynomials, such as the $R_{s,v}$ of Theorem 1.1, in the above setting is the most studied of its kind. Indeed, Y. André, see [And95] and F. Beukers, see [Beu93], have already given independently, and with different methods, partial answers to this.

In more detail, F. Beukers constructed in [Beu93], under assumptions on the morphism $f : \mathcal{E} \to S$, polynomials $R_{s,v}$ as in Theorem 1.1 when v is an archimedean place of $K(s)$ and when v is finite place with $v \nmid 2, 3$. On the other hand, Y. André in [And95] constructed polynomials $R_{s,v}$ as in Theorem 1.1 under the assumptions that

1. v is a place of supersingular reduction of the "central fiber" $\mathcal{E}_{s_0}$ and

2. v is over some prime $p \in \mathbb{Q}$ which is unramified in the CM field $\mathrm{End}^0_{\bar{\mathbb{Q}}}(\mathcal{E}_{s_0})$.

Following the work of André and Beukers, the truly new $R_{s,v}$ in Theorem 1.1 are those for places v such that

1. v is over some prime $p \in \mathbb{Q}$ which ramifies in $\mathrm{End}^0_{\bar{\mathbb{Q}}}(\mathcal{E}_{s_0})$, or

2. v is a place over which the central fiber $\mathcal{E}_{s_0}$ has ordinary reduction.

A striking new feature of our construction is that, for v over which $\mathcal{E}_{s_0}$ has ordinary reduction, the relations we construct are independent of the place v. More precisely, we obtain a single polynomial $R_{s,\mathrm{ord}}$, for which $\iota_v(R_{s,\mathrm{ord}}(Y_G(s))) = 0$ for all such places for which s and s_0 are v-adically close. In particular, this polynomial can be constructed from any one such place v and does not depend on the number of such places. Equivalently, the coefficients of $R_{s,\mathrm{ord}}$ are independent of v. This phenomenon appears to reflect a previously unnoticed structural property of the G-functions method, and persists in all other cases considered in this series of papers.

1.2 Height bounds and applications

The modern study of the values of G-functions at points of "special interest" has its beginnings at the work of E. Bombieri, see [Bom81], and Y. André, see [And89]. The ultimate goal of the "G-functions method", developed by Y. André in [And89], is to establish height bounds for such points of special interest.

In the geometric setting described in Theorem 1.1 the height bounds one gets are of the form

$$h(s) << \deg(\prod_{v} R_{s,v})^{c_0}, \tag{1}$$

where the product runs over all places in $K(s)$ for which s is v-adically close to s_0.

These height bounds are connected with Siegel's lower bounds for the class numbers of imaginary quadratic number fields. Indeed, were we to know that the degree of the polynomial on the right hand side of (1) was bounded polynomially by $[\mathbb{Q}(s) : \mathbb{Q}]$, we would have an effective version of Siegel's bounds in [Sie35] upon combining our height bound with results of Masser-Wüstholz. For more on this we refer the reader to the introduction of [Pap26].

While being unable to obtain such an upper bound for the aforementioned degree, in Section 6 we discuss the implications of our relations to the search for an effective version of Siegel's result from the point of view of the G-functions method. In short, we show that the problem may be reduced to finding a bound for the size of the subset $\mathcal{P}(s, s_0)$ of $\Sigma_{K(s)}$ that consists of places v for which

1. s is v-adically close to s_0 and
2. $\mathcal{E}_{s_0}$ has supersingular reduction over v and
3. v is over some prime $p \in \mathbb{Q}$ which is unramified in $\mathrm{End}^0_{\mathbb{Q}}(\mathcal{E}_{s_0})$.

Alternatively, we would like to have a measure of dependence on v for precisely those $R_{s,v}$ in Theorem 1.1 that were constructed by Y. André in [And95].

1.3 Outline of the paper

We start in Section 2 with some foundational results on the periods of CM elliptic curves, either the archimedean periods coming from the de Rham-Betti comparison isomorphism or the non-archimedean ones coming from the

de Rham-crystalline comparison isomorphism. Following this, in Section 3 we briefly review the basic framework through which one may associate a family of G-functions to a 1-parameter family of elliptic curves.

Next, in Section 4 we discuss briefly a generalization of [GZ85], due to Lauter-Viray [LV15], and record an application relevant to places of proximity over ramified primes of the CM field of endomorphisms of the central fiber. Section 5 consists of a proof of Theorem 1.1 working on a case-by-case basis. More specifically, we describe the v-adic relations for v a place of ordinary reduction of the "central fiber" $\mathcal{E}_{s_0}$ of our family of elliptic curves. We also record here the relations of André, in the supersingular case, and Beukers, in the archimedean setting. We conclude this section by describing the relations for places over primes ramified in the field $\mathrm{End}^0_{\bar{\mathbb{Q}}}(\mathcal{E}_{s_0})$.

Finally, in Section 6, we discuss the height bounds that we alluded to in Section 1.2. We conclude our exposition by discussing the implications of our height bounds to the search for an effective version of Siegel's result.

1.4 Notation

Let E be an elliptic curve over a number field K and $v \in \Sigma_K$ a place of K. We write E_v for the base change $E \times_K K_v$. If v is a finite place of good reduction we will write $\tilde{E}_v$ for the reduction of E modulo v.

Given a family of power series $\mathcal{Y} := (y_1, \ldots, y_N) \in K[[x]]$ where K is as above, and $v \in \Sigma_K$ is some place of K, we will write $R_v(y_j)$ for the v-adic radius of convergence of y_j. We also set $R_v(\mathcal{Y}) := \min R_v(y_j)$. Given v as above, we write $\iota_v : K \hookrightarrow \mathbb{C}_v$ for the associated embedding into $\mathbb{C}_v$, which will stand for either $\mathbb{C}$ or $\mathbb{C}_p$ depending on whether the place v is archimedean or not. Finally, if $y(x) = \sum_{n=0}^{\infty} a_n x^n \in K[[x]]$ is a power series as above we will write $\iota_v(y(x)) := \sum_{n=0}^{\infty} \iota_v(a_n) x^n$ for the corresponding power series in $\mathbb{C}_v[[x]]$.

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2 Period matrices of CM elliptic curves

We start in this section with some general facts about the arithmetic of our main objects of study, CM elliptic curves and their periods.

Associated to any elliptic curve E defined over a number field K we will get, for each place $v \in \Sigma_K$, a certain period matrix. This will be nothing but the matrix associated to a comparison isomorphism between two cohomology theories, either de Rham and Betti or de Rham and crystalline, and some bases for the respective cohomology groups. In the sequel it will be useful for us to work with particular bases of $H^1_{dR}(E/K)$. We chose to give these the following:

Definition 2.1. *Let E be an elliptic curve defined over a number field K. We call an ordered basis $\Gamma_{dR}(E) := \{\omega, \eta\}$ of $H^1_{dR}(E/K)$ a **Hodge basis** if the following are true:*

1. *$\omega \neq 0$ is in the first part of the filtration $F^1 = e^*\Omega_{E/K} \subset H^1_{dR}(E/K)$, and*

2. *η is such that $\langle \omega, \eta \rangle = 1$ with respect to the Riemann bilinear form.*

2.1 Canonical Crystalline bases

For the remainder of this section we let E/K be a CM elliptic curve defined over a number field K and let $v \in \Sigma_K$ be a place of K. We will work throughout under the further assumptions[1]that E has everywhere good reduction, that $F := \mathrm{End}^0_{\bar{\mathbb{Q}}}(E) = \mathrm{End}^0_K(E)$, where $[F : \mathbb{Q}] = 2$ is the CM field of endomorphisms of E, and that $F \subset K$.

As per usual notational conventions in p-adic Hodge theory we let $K_{v,0} := W(k_v)[\frac{1}{p}]$, where k_v is the residue field of K at v and p stands for its characteristic. We also denote by $q := p^f$ the cardinality of k_v. From now on, we

[1]These can be achieved upon base changing E by a finite extension L/K. The "everywhere good reduction" condition follows from [ST68]. We also note here that the assumption that $\mathrm{End}^0_K(E) = \mathrm{End}^0_{\bar{\mathbb{Q}}}(E)$ is equivalent to the assumption that $F \subset K$, see for example the Remark at the beginning of §10 in [Lan87].

will also write

$$H^1_v(E) := \begin{cases} H^1_{\text{crys}}(\tilde{E}_v/W(k_v)) \otimes K_{v,0} & \text{if } v \in \Sigma_{K,f} \text{ and} \\ H^1_B(E^{an}_v, \mathbb{Q}) & \text{if } v \in \Sigma_{K,\infty}. \end{cases} \tag{2}$$

Over finite places v, since E has good reduction at v, we have canonical isomorphisms, which we simply refer to as the "de Rham-crystalline comparison isomorphism",

$$\rho_v(E) : H^1_{dR}(E_v/K_v) \to H^1_v(E) \otimes_{K_{v,0}} K_v. \tag{3}$$

These are due to Berthelot-Ogus, see Theorem 2.4 of [BO83]. This mirrors the classical "de Rham-Betti" comparison of Grothendieck, for $v \in \Sigma_{K,\infty}$ this time, which we will denote again by

$$\rho_v(E) : H^1_{dR}(E/K) \otimes_K \mathbb{C} \to H^1_v(E) \otimes_{\mathbb{Q}} \mathbb{C}. \tag{4}$$

Our goal in this subsection is to associate to each $v \in \Sigma_{K,f}$ as above a "canonically" chosen symplectic basis of $H^1_v(E)$, denoted $\Gamma_v(E)$ in the sequel. We shall do this by working in cases depending on the reduction of E at the place v.

Definition 2.2. *Let E/K be a CM elliptic curve as above and $v \in \Sigma_{K,f}$ a finite place of K. We say that v is an **ordinary** (resp. **supersingular**) **place for** E if the reduction $\tilde{E}_v$ of E at v is an ordinary (resp. supersingular) elliptic curve over the finite field k_v.*

2.1.1 Ordinary reduction

We start with the case where v is an ordinary place for E. Let us also write $F_v \in \text{End}(\tilde{E}_v)$ for the q-th power Frobenius endomorphism of the reduction at v, where $q = p^f$ is as above. We write ϕ_v for the endomorphism of the crystalline object $H^1_v(E)$ induced by F_v via pullback.

Since $\tilde{E}_v$ is ordinary we know that the slopes of ϕ_v are 1 and 0. In the case of Dieudonné modules this is classical, see for example the discussion on page 98 of [Dem72]. To deduce this about the action of ϕ_v on H^1_{crys} one needs the comparison between H^1_{crys} and the Dieudonné module of $\tilde{E}_v$ of Berthelot-Breen-Messing, see [BBM82].

Letting λ_0, λ_1 denote the eigenvalues of ϕ_v we know that these will satisfy $v(\lambda_i) = i$, see for example Manin's theorem on page 98 of [Dem72]. Now we choose γ_v, resp. δ_v, to be a non-zero eigenvector of ϕ_v with eigenvalue λ_1, resp. λ_0. These will necessarily be linearly independent, since $\lambda_0 \neq \lambda_1$. Therefore for the Riemann form $\langle \gamma_v, \delta_v \rangle = \mu \neq 0$. Rescaling one of these vectors, say δ_v, by μ^{-1} we get a basis of $H^1_v(E)$ that is symplectic.

Definition 2.3. *We let* $\Gamma_v(E) := \{\gamma_v, \delta_v\}$ *be the above symplectic basis of* $H^1_{\mathrm{crys}}(\tilde{E}_v/W(k_v)) \otimes K_{v,0}$ *and call it a* ***canonical basis*** *of* $H^1_v(E)$.

The basis $\Gamma_v(E)$ chosen above is obviously not unique, but rather unique up to a scalar. Nevertheless, this basis enjoys a certain "canonicity for morphisms", due to its construction reflecting the action of Frobenius. In more detail, we have the following:

Lemma 2.4. *Let* $\tilde{E}$ *and* $\tilde{E}'$ *be ordinary elliptic curves over* k_v *and let* $f \in \mathrm{Hom}(\tilde{E}, \tilde{E}')$ *be a non-zero homomorphism. Let* $\Gamma_v(\tilde{E}) = \{\gamma, \delta\}$ *and* $\Gamma_v(\tilde{E}') = \{\gamma', \delta'\}$ *be chosen as above.*

Then, for its pullback $f_{\mathrm{crys}} : H^1_{\mathrm{crys}}(\tilde{E}'/K_{v,0}) \to H^1_{\mathrm{crys}}(\tilde{E}/K_{v,0})$, *there exist* $\zeta_0, \zeta_1 \in K_{v,0}$ *such that* $f_{\mathrm{crys}}(\gamma') = \zeta_1 \cdot \gamma$ *and* $f_{\mathrm{crys}}(\delta') = \zeta_0 \cdot \delta$.

Proof. The pullback f_{crys} of f on the level of crystalline cohomology will be an endomorphism in the category of F-isocrystals. In other words, we have $f_{\mathrm{crys}} \circ \phi'_{\mathrm{crys}} = \phi_{\mathrm{crys}} \circ f_{\mathrm{crys}}$, where ϕ'_{crys} stands for the q-th power Frobenius of $H^1_{\mathrm{crys}}(\tilde{E}'/K_{v,0})$.

Applying this to γ' it is easy to see that we must have that $\phi_{\mathrm{crys}}(f_{\mathrm{crys}}(\gamma')) = \lambda_1 \cdot f_{\mathrm{crys}}(\gamma')$. Therefore, since ϕ_{crys} is linear with distinct eigenvalues, we get that there exists $\zeta_1 \in K_0$ such that $f_{\mathrm{crys}}(\gamma') = \zeta_1 \cdot \gamma$. The existence of ζ_0 for δ follows similarly. □

2.1.2 Supersingular reduction

Here we assume that E has supersingular reduction at the finite place v, which we consider fixed for now. Throughout Section 2.1.2, with Lemma 2.6 in mind, we assume that the prime $p := \mathrm{char}(k_v)$ is unramified in the quadratic field $F = \mathrm{End}^0(E)$.

The description of the basis in this case is due to Y. André in [And95], see in particular §5. For a more in depth description we point the interested reader to §5.3.1 in [And03] and Proposition 5.3.3 and the subsequent remark there. For the reader's convenience, we include here a summary of this construction of André.

Let us consider the endomorphism algebra $\mathrm{End}^0_{k_v}(\tilde{E}_v)$. Since we know that the reduction at v is supersingular, the algebra $\mathrm{End}^0_{\bar{k}_v}(\tilde{E}_v)$ will be the unique quaternion algebra $D_{p,\infty}$ over $\mathbb{Q}$ that is ramified only at p and ∞. In particular we may find a finite extension $\mathbb{F}_{q'}/\mathbb{F}_q$ for which $\tilde{E}' := \tilde{E}_v \times_{\mathbb{F}_q} \mathrm{Spec}(\mathbb{F}_{q'})$ is such that $\mathrm{End}^0_{\mathbb{F}_{q'}}(\tilde{E}') = D_{p,\infty}$.

Writing $K'_{v,0}/K_{v,0}$ for the unique unramified extension with residue field $\mathbb{F}_{q'}$ we then get that $D_{p,\infty}$ acts on the crystalline cohomology $H^1_{\mathrm{crys}}(\tilde{E}'/K'_{v,0}) =$

$H^1_v(E) \otimes_{K_{v,0}} K'_{v,0}$. Tensoring both sides of the canonical isomorphism (3) of Berthelot-Ogus by the compositum $K'_v := K'_{v,0}K_v$ we get

$$H^1_{dR}(E/K) \otimes_K K'_v \to (H^1_v(E) \otimes_{K_{v,0}} K'_{v,0}) \otimes_{K'_{v,0}} K'_v. \tag{5}$$

Since we assumed that p is unramified in F we know that $F \subset K_{v,0}$. Furthermore, since $F \hookrightarrow D_{p,\infty}$ we know that $D_{p,\infty} \otimes_{\mathbb{Q}} F \simeq M_2(F)$, see for example Proposition 1.2.3 in [GS06]. Fixing the embedding $F \hookrightarrow \mathbb{C}_v$ induced from that of K_v, as discussed in $5.3.2 - 5.3.3$ in [And03], this leads to a canonical F-subspace $H^1_B(\tilde{E}, F) := M_2(F) \cdot \gamma_v \subset H^1_{\text{crys}}(\tilde{E}_v \times_{\bar{\mathbb{F}}_q})$ where $\gamma_v \in H^1_{\text{crys}}(\tilde{E}'/K'_{v,0})$ is some eigenvector for the action of F viewed as a sub-algebra of $D_{p,\infty}$ now.

One may then find a $\delta_v \in H^1_B(\tilde{E}, F)$ which is linearly independent to γ_v and such that $\langle \gamma_v, \delta_v \rangle = 1$ as explained in [And95]. As in [And95] we can furthermore pick δ_v so that it is a complementary eigenvector for the action of F. In other words, if say $F = \mathbb{Q}(\sqrt{-d})$ we may assume that $e_{\text{crys}}(\gamma_v) = \sqrt{-d} \cdot \gamma_v$ and $e_{\text{crys}}(\delta_v) = -\sqrt{-d} \cdot \delta_v$, where e_{crys} denotes the pullback on the level of crystalline cohomology of $e := \sqrt{-d} \in F$, viewed as an element of the endomorphism algebra of our elliptic curve.

Definition 2.5. *We call the above pair* $\Gamma_v(E) := \{\gamma_v, \delta_v\}$ *a* ***canonical basis*** *of* $H^1_v(E)$.

2.2 Period relations

The main period relations, in either the archimedean or p-adic setting, among the periods of a CM elliptic curve are summarized in the following:

Lemma 2.6. *Let* E, F *be as above and let* $v \in \Sigma_K$ *be a place of* K. *Then there exists a Hodge basis* $\Gamma_{dR}(E)$ *of* $H^1_{dR}(E/K)$ *such that*

1. *if* $v \in \Sigma_{K,\infty}$, *there exists* $\varpi_v \in \mathbb{C}_v$, *such that ,with respect to* $\Gamma_{dR}(E)$ *and a choice of a symplectic basis of* $H^1(E^{an}_v, \mathbb{C})$, *the period matrix of* E *is of the form*
$$\begin{pmatrix} \frac{\varpi_v}{2\pi i} & 0 \\ 0 & \varpi_v^{-1} \end{pmatrix}. \tag{6}$$

2. *if* $v \in \Sigma_{K,f}$ *is a finite place of good reduction of* E *for which* $p = \text{char}(k_v)$ *is unramified in* $F/\mathbb{Q}$, *there exists* $\varpi_v \in \mathbb{C}_v$ *such that, with respect to* $\Gamma_{dR}(E)$ *and a choice of a symplectic basis of* $H^1_{\text{crys}}(\tilde{E}_v/W(k_v)) \otimes \mathbb{C}_v$, *the period matrix of* E *is of the form*
$$\begin{pmatrix} \varpi_v & 0 \\ 0 & \varpi_v^{-1} \end{pmatrix}. \tag{7}$$

Remark 2.7. *We believe that the assertions here are classical and known to experts in either setting. For instance, these are used in passing by Y. André in [And95] in the archimedean case and the case of primes of supersingular reduction of E. For a rather detailed proof in the archimedean case we point the interested reader to [Pap26], Lemma 2.9. We present here a similar argument in the case of finite places.*

Proof. Let $V_{dR} := H^1_{dR}(E/K)$ and $H^1_v(E) := H^1_{\text{crys}}(\tilde{E}_v/K_{v,0})$, as per our usual notation. Let us also write $F := \mathbb{Q}(\sqrt{-d})$ with $d \in \mathbb{Z}$ square-free and let $e_{dR} \in \text{End}(V_{dR})$ and $e_{\text{crys}} \in \text{End}(H^1_v(E))$ denote the elements corresponding to the action of $e = \sqrt{-d} \in F$ in the de Rham and crystalline cohomology groups respectively.

We then get for V_{dR} and $H^1_v(E)$ splittings of the form

$$V_{dR} = V^+_{dR} \oplus V^-_{dR}, \text{ and} \tag{8}$$

$$H^1_v(E) := W^+_v \oplus W^-_v, \tag{9}$$

induced by the action of F. Here V^+_{dR} (resp. W^+_v) denotes the subspace on which e_{dR} (resp. e_{crys}) acts as multiplication by $\sqrt{-d}$ and V^-_{dR} (resp. W^-_v) the subspace on which this action is multiplication by $-\sqrt{-d}$. Note here that in order to have this splitting into proper subspaces of $H^1_v(E)$ it is essential that the prime p is unramified in $F/\mathbb{Q}$. Indeed, the assumption $F \leq K$ paired with the fact that p is unramified implies that $F \leq K_{v,0}$, where $K_{v,0}/\mathbb{Q}_p$ is now unramified by definition.

Let us write $\rho_v := \rho_v(E)$ for the de Rham-crystalline isomorphism of [BO83] and let ω, η be non-zero vectors in V^+_{dR}, V^-_{dR} respectively, as well as γ, δ non-zero vectors in W^+_v and W^-_v respectively. The linear independence of these pairs guarantees that we may furthermore, which we do from now on, assume that they form symplectic bases of V_{dR}, and $H^1_v(E)$ respectively. To see this note that the linear independence of, say, ω, η guarantees that $\langle \omega, \eta \rangle = \alpha \neq 0$, where $\langle \cdot, \cdot \rangle$ denotes the Riemann form on V_{dR}, at which point we may replace η by $\alpha^{-1} \cdot \eta$.

Let us write $(a_{i,j})$ for the period matrix corresponding to the choice of these bases via ρ_v. By compatibility of the action of F on H^1_{dR} and H^1_{crys} with ρ_v we get that

$$\begin{aligned} &\alpha_{1,1}(\sqrt{-d})\gamma + \alpha_{1,2}(\sqrt{-d})\delta = \rho_v(e_{dr} \cdot \omega) = \\ &= e_{\text{crys}} \cdot \rho_v(\omega) = \alpha_{1,1}(\sqrt{-d})\gamma + \alpha_{1,2}(-\sqrt{-d})\delta, \end{aligned}$$

and similarly for the action of e on η. Comparing coefficients we see that we must have $\alpha_{i,j} = 0$ if $i \neq j$, so that $(\alpha_{i,j})$ is diagonal.

The algebraicity of the construction of the Riemann form and its compatibility with the de Rham-crystalline comparison, see Chapter 5 in [BBM82],

gives that $\langle \rho_v(\omega), \rho_v(\omega') \rangle = \langle \omega, \omega' \rangle$ for all ω, ω'. Applying this with ω and η shows that $\alpha_{1,1}\alpha_{2,2} = 1$ thus concluding the proof. □

Remarks 2.8. *1. The basis $\Gamma_{dR}(E)$ does not depend on the place v and is the same as that chosen in the proof of Lemma 2.9 in [Pap26], i.e. the archimedean case of part 1 of the previous lemma.*

2. The "canonical" bases chosen in the discussion preceding Definition 2.3, in the case of ordinary reduction of E, resp. the discussion right before Definition 2.5 in the case of supersingular reduction, can be used in the proof of Lemma 2.6, at least up to permutation.

To see this, in the case of ordinary reduction, note that by Lemma 2.4 we will have that γ_v and δ_v will be eigenvectors of e_{crys} in the notation of the proof. Note that if $e_{\text{crys}}(\gamma_v) = -\sqrt{d} \cdot \gamma_v$ then we let $\gamma := -\delta_v$ and $\delta := \gamma_v$, where the "$-$" is needed to ensure that $\langle \gamma, \delta \rangle = 1$.

In the case of supersingular reduction this follows by construction of the basis in [And95], see for example §5.c there for more details.

2.3 Reductions of CM elliptic curves

Let E, defined over a number field K, be a CM elliptic curve as in Section 2.1. The description of the type of reduction of E at places of K is classically known here and is due to Deuring. For a proof we point the interested reader to [Lan87] §13., Theorem 12.

Phrased under our assumptions in Section 2.1 this is the following:

Theorem 2.9 (Deuring's Reduction Theorem)**.** *Let E/K be a CM elliptic curve that has everywhere good reduction and is such that $F := \text{End}^0_K(E) = \text{End}^0_{\overline{\mathbb{Q}}}(E)$. Let $v \in \Sigma_{K,f}$ be a finite place of K and let $p \in \mathbb{Q}$ be the prime over which v lies.*

The reduction $\tilde{E}_v$ of E at v is supersingular if and only if p either ramifies or is inert in F.

3 Background on the G-functions method

Here we summarize the connection between G-functions and a 1-parameter family of elliptic schemes. For a more detailed exposition, we point the interested reader to §3 of [Pap25a], whose terminology we have chosen to follow, as well as [DOP25].

The geometric picture permeating the rest of the paper is the one outlined in Theorem 1.1. Namely, we consider some 1-parameter family of elliptic

curves $f : \mathcal{E} \to S$ defined over a number field K, together with a distinguished point $s_0 \in S(K)$ whose fiber is a CM elliptic curve, which we denote by E_0 for ease of notation. From the perspective of the G-functions method it is convenient to make assumptions about both the "central fiber" E_0 and the curve S, as long as those may be attained by base changing our original picture by either a finite cover S' of S or by tensoring the entire picture by a finite extension K'/K.

With this in mind, we assume from now on that E_0 has everywhere good reduction, see [ST68], and that all endomorphisms of E_0 are defined over the base field K, i.e. that $\mathrm{End}^0_{\bar{\mathbb{Q}}}(E_0) = \mathrm{End}^0_K(E_0)$ is some CM field.

In this context, of allowing base changes to the original picture, one may find a "local parameter" $x \in K(S)$ at s_0 and a tuple of power series $\mathcal{Y} \in \bar{\mathbb{Q}}[[x]]^n$ that one can think of as "power series on the curve S centered at s_0". These power series appear, in a canonical way, as the entries of matricial solution to the system of differential equations one gets by considering the differential module $(H^1_{dR}(\mathcal{E}/S), \nabla)$ together with a basis of sections of this module on an affine neighborhood of s_0.

In our setting $\mathcal{Y}$, often denoted $Y_G(x)$ in the sequel, is nothing but a collection of matrices in $\mathrm{GL}_2(\bar{\mathbb{Q}}[[x]])$. In practice, the number of such matrices depends on the preimages of our original s_0 under a potentially necessary finite cover $S' \to S$, see [DO25] where these ideas were first introduced. For reasons of expositional simplicity we have chosen to consider the case where there is only one such matrix in our family, especially since this is the crucial step in the method we employ.

Considering the values of $Y_G(x)$ at points $s \in S(\bar{\mathbb{Q}})$ makes sense only after we have introduced some topology to our curve, which we do via considering the analytification of our curve S, either in the complex analytic or p-adic sense, with respect to some place v of K. The v-adic values $\iota_v(Y_G(x(s)))$ then make sense as long as $|x(s)|_v$ is smaller than the v-adic radius of convergence of the family $Y_G(x)$. We thus naively think of v-adic analytic discs Δ_v centered at our s_0 on the curve with sufficiently small radii as part of setting.

For finite places v, after possibly altering the family Y_G, we may select Δ_v so that if $s \in S(\bar{\mathbb{Q}}) \cap \Delta_v$ then $\mathcal{E}_s$ and E_0 have the same reduction modulo v. For more on this, see §3.3.2 of [Pap25a]. With this further property for the Δ_v in mind for finite places, from now on we adopt the following:

Definition 3.1. *Let $s \in S(\bar{\mathbb{Q}})$ and $w \in \Sigma_{K(s)}$ and $v \in \Sigma_K$ such that $w|v$. We say that s is w-adically close to s_0 if $s \in \Delta_v$ for the analytic discs chosen above.*

On such Δ_v the values of our G-functions attain cohomological signifi-

cance. In the archimedean setting this is due to work of André in [And89] and in the p-adic setting the first observation of this phenomenon appears in André's [And95]. In short, we have in our disposal relative versions of the comparison isomorphisms that appear in Section 2. In the archimedean setting this is classically due to Grothendieck and in the p-adic setting this is due to work of Berthelot-Ogus, see [BO83], and Ogus, see [Ogu84], in the ramified case. All in all, for $s \in \Delta_v \cap S(\bar{\mathbb{Q}})$ we will have

$$\mathcal{P}_v(s) = \iota_v(Y_G(x(s)))\mathcal{P}_v(s_0), \tag{10}$$

where $\mathcal{P}_v$ stand for the matrices associated to the aforementioned isomorphism, the choice of a basis of sections of $H^1_{dR}(\mathcal{E}/S)$, and a basis of the fiber $H^1_v(E_0)$, which we may identify with the horizontal sections of $(H^1_{dR}(\mathcal{E}/S), \nabla)$ on the disc Δ_v. For more details on this we point the interested reader to the proof of Theorem 3.4 in [Pap25a].

In constructing relations among the v-adic values of the $Y_G(x)$ at points of interest, we will want to make certain that these are "non-trivial" in the terminology of $VII.5$ in [And89]. Our main tool in this direction is the description of the "trivial relations" among the members of the family $Y_G(x)$. We record these in the following:

Proposition 3.2. *Let $Y_G(x)$ be the family of G-functions associated to $(\mathcal{E} \to S, s_0)$ as above. Then $Y_G(x)^{\mathrm{Zar}} \subset (M_{2\times 2})_{\bar{\mathbb{Q}}(x)}$ is the subvariety cut out by the ideal*

$$I(\mathrm{SL}_2) = \{\det(X_{i,j}) - 1\}, \tag{11}$$

where $X_{i,j}$ are such that $M_{2\times 2}$ is given by $\mathrm{Spec}(\mathbb{Q}[X_{i,j} : 1 \leq i, j \leq 2])$.

This result seems to be classical, it is mentioned without proof in [Beu93] for example. For a proof we point the interested reader to §4.5.6 of [DOP25].

4 p-adic proximity

The issue of "p-adic proximity" of s to s_0 may be linked directly to the work of Gross-Zagier in [GZ85]. We record here, for the convenience of the reader, a generalization of their results due to Lauter-Viray. As a consequence of this we conclude with a short lemma we will need in the sequel around p-adic proximity of CM points to our center s_0, for primes p ramified in the CM field giving the algebra of endomorphisms of the "central fiber" E_0 in the notation of Section 3.

4.1 A formula of Gross-Zagier

Let d_j, for $j = 1, 2$, be the discriminants of two distinct quadratic imaginary orders $\mathcal{O}_j$ and consider the product

$$J(d_1, d_2) := \prod_{\substack{[\tau_1],[\tau_2] \\ \mathrm{disc}(\tau_j)=d_j}} (j(\tau_1) - j(\tau_2))^{\frac{8}{w_1 w_2}}, \tag{12}$$

with $[\tau_j]$ ranging through all elements of $\mathcal{H}/\operatorname{SL}_2(\mathbb{Z})$ with prescribed discriminants and where w_j denote the number of roots of unity in the order $\mathcal{O}_j$.

In [GZ85], Gross and Zagier established a remarkable formula for the quantity $J(d_1, d_2)$ under the assumptions that

1. d_1 and d_2 are relatively prime and
2. the d_j are fundamental discriminants, i.e. discriminants of maximal orders.

This formula of Gross-Zagier was further generalized, fairly recently by Lauter-Viray, see [LV15] Theorem 1.1, to the case where $d_1 \neq d_2$ and without any assumptions on the maximality of the corresponding orders.

As a corollary of their formula, see [GZ85] Corollary 1.6, Gross-Zagier get an elegant description of the primes dividing $J(d_1, d_2)$, albeit under their aforementioned assumptions on the pair (d_1, d_2). The natural generalization of this, again due to Lauter-Viray, is the following:

Theorem 4.1 (Lauter-Viray, [LV15] Corollary 1.3)**.** *Let $d_1 \neq d_2$ be two distinct discriminants as above and let $l \in \mathbb{N}$ be a prime with $l | J(d_1, d_2)$. Then*

1. *$d_2 = d_1 \cdot l^{2k}$ for some $k \in \mathbb{Z}\backslash\{0\}$, or*
2. *there exists an integer $m = \frac{d_1 d_2 - x^2}{4} > 0$ with $l|m$ such that the Hilbert symbol $(d_j, -m)_p \neq 1$ if and only if $p = l$.*

In particular, if $l \nmid d_1 d_2$ then $\left(\frac{d_1}{l}\right) = \left(\frac{d_2}{l}\right) = -1$.

4.2 Application to primes of proximity

Here we consider the geometric picture sketched in Section 3. Namely we consider a fixed 1-parameter family $f : \mathcal{E} \to S$ of elliptic curves defined over a number field K. As in Section 3 we assume that we are also given a point $s_0 \in S(K)$ for which $E_0 := \mathcal{E}_{s_0}$ is a CM elliptic curve with everywhere good reduction and with $F_0 := \operatorname{End}^0_{\bar{\mathbb{Q}}}(E_0) = \operatorname{End}^0_K(E_0)$.

In the above picture, following Definition 3.1 and the conventions around it, we have:

Lemma 4.2. *Let $s \in S(\bar{\mathbb{Q}})$ be such that $\mathcal{E}_s$ is a CM elliptic curve and write $d_P = \mathrm{disc}(\mathrm{End}_{\bar{\mathbb{Q}}}(\mathcal{E}_P))$, for $P \in \{s_0, s\}$, to be the corresponding discriminant. Let us also fix a finite place $v \in \Sigma_{K,f}$ over a rational prime p, where p is ramified in the CM field $F_0 = \mathrm{End}^0(E_0)$.*

If s is v-adically close to s_0 then $\mathrm{End}^0_{\bar{\mathbb{Q}}}(\mathcal{E}_s) = F_0$, i.e. $\mathcal{E}_s$ has CM by the same field as E_0.

Proof. Assume that it is not so. In particular, we must have that $d_s \neq d_{s_0}$.

By our conventions on the disks Δ_v we get that $\mathcal{E}_s$ and E_0 have the same reduction over a place $w \in \Sigma_{K(s),f}$ with $w|v$. In particular, $w(j(\mathcal{E}_s) - j(E_0)) > 0$ and hence $p|J(d_s, d_{s_0})$. Note that since p is ramified in F_0, we must have $p|d_{s_0}$. Now we apply Theorem 4.1 to conclude that $d_s = p^{2k} d_{s_0}$ for some $k \in \mathbb{Z}$ and our result follows. □

5 Relations among values of G-functions

This constitutes the main technical part of our text. In short, we describe the construction of relations among the values of G-functions associated to a 1-parameter family $f : \mathcal{E} \to S$ of elliptic curves at points $s \in S(\bar{\mathbb{Q}})$ whose corresponding fibers are CM elliptic curves.

In essence, this section is a proof of Theorem 1.1. The construction of the relations described here depends on the "type" of the place v with respect to the central fiber E_0.

5.1 Notation

Throughout this section we work in a somewhat simplified form of the general setting described in the previous section. For the remainder of this section we consider fixed a 1-parameter family of elliptic curves $f : \mathcal{E} \to S$, where S is a smooth and geometrically irreducible curve over a number field K. We also consider a fixed point $s_0 \in S(K)$ for which we assume that the fiber is a CM elliptic curve $E_0 := \mathcal{E}_{s_0}$.

In order to simplify our exposition, throughout this section we assume that the curve E_0 satisfies the following:

1. E_0 has everywhere[2] good reduction,

[2] A fact we may assume up to base changing our original morphism π by some finite extension of K due to [ST68].

2. $\mathrm{End}_{\bar{\mathbb{Q}}}(E_0) = \mathrm{End}_K(E_0) = F_0$, and

3. there exists a uniformizer $x \in K(S)$ that has a simple zero at s_0.

For the remainder of this subsection we also fix a place $v \in \Sigma_K$ as well as a point $s \in S(K)$ which is v-adically close to s_0 in the sense discussed in Section 3.

Given the above data, we choose a symplectic basis $\{\omega, \eta\}$ of $H^1_{dR}(\mathcal{E}/S)$ whose fiber at s_0 consists of F_0-eigenvectors, as in the proof of Lemma 2.6. We also may choose ω so that it is a section of $F^1H^1_{dR}(\mathcal{E}/S)$. This does not interfere with the assumption that the fiber ω_0 is an eigenvector for the action of F_0, since $F^1H^1_{dR}(E_0/K)$ are precisely the invariant differentials. From Section 3 we therefore get a matrix $Y \in \mathrm{SL}_2(\bar{\mathbb{Q}}[[x]])$ of G-functions associated with this choice of basis.

5.2 Relations at ordinary places

We start with the case where the place v is a finite place of ordinary reduction for the "central fiber" E_0. In this case we have:

Proposition 5.1. *With notation as in Section 5.1, let $Y_G = (Y_{i,j}(x)) \in \mathrm{SL}_2(\bar{\mathbb{Q}}[[x]])$ be the matrix of G-functions associated to the pair (f, s_0) as in Section 3.*

Then for all $s \in S(\bar{\mathbb{Q}})$ for which $\mathrm{End}^0_{\bar{\mathbb{Q}}}(\mathcal{E}_s) = \mathrm{End}^0_{\bar{\mathbb{Q}}}(E_0)$ and for all finite places of $\Sigma_{K(s)}$ that are ordinary for E_0 and are such that s is v-adically close to s_0 we have $\iota_v(Y_{1,2}(x(s))) = 0$.

Proof. We begin by noting that our assumption that E_0 has ordinary reduction at v allows us to use Lemma 2.6. Indeed by Deuring's criterion, see Theorem 12 in §13.4 of [Lan87], having ordinary reduction implies that the prime $p = \mathrm{char}\, k_v$ is unramified in F_0.

We choose a basis $\{\gamma, \delta\}$ of $H^1_{\mathrm{crys}}(\tilde{E}_{0,v}/K_{0,v})$ as in Lemma 2.4. Then, again by Lemma 2.6, since both this as well as the chosen basis $\{\omega_0, \eta_0\}$ of $H^1_{dR}(E_0/K)$ are symplectic and consist of F_0-eigenvectors in the respective spaces, the matrix of relative periods associated to the comparison isomorphism $\mathcal{P}_v$ will be of the form $\mathcal{P}_v(s) = \iota_v(Y(x(s))) \cdot \begin{pmatrix} \varpi_v & 0 \\ 0 & \varpi_v^{-1} \end{pmatrix}$, where $\varpi_v \in \mathbb{C}_v$.

Since $\mathcal{E}_s$ and E_0 have CM by the same CM-field there exists an isogeny $\phi : \mathcal{E}_s \to E_0$. We write ϕ_{dR} and ϕ_{crys} for the morphisms induced from ϕ in the respective cohomology theories.

Functoriality in the de Rham-crystalline comparison implies that

$$\mathcal{P}_v(s) \circ \phi_{dR} = \phi_{\text{crys}} \circ \mathcal{P}_v(0). \tag{13}$$

Note also that ϕ_{crys} is induced from the reduction $\tilde{\phi}_v \in \text{End}(\tilde{E}_{0,v})$ of ϕ. Thus by Lemma 2.4 we get that there exist ζ_v, ξ_v such that $[\phi_{\text{crys}}] = \begin{pmatrix} \zeta_v & 0 \\ 0 & \xi_v \end{pmatrix}$ for the matrix of ϕ_{crys} with respect to the basis of $H^1_v(E_0)$ fixed above.

Writing $\phi_{dR}\omega_0 = a \cdot \omega_s$ and $\phi_{dR}\eta_0 = c \cdot \omega_s + d \cdot \eta_s$ we may thus translate (13) into the following equation between period matrices

$$\begin{pmatrix} \iota_v(a) & 0 \\ \iota_v(c) & \iota_v(d) \end{pmatrix} \mathcal{P}_v(s) = \mathcal{P}_v(0) \begin{pmatrix} \zeta_v & 0 \\ 0 & \xi_v \end{pmatrix} \tag{14}$$

Setting $y_{i,j} := \iota_v(Y_{i,j}(x(s)))$, since $\mathcal{P}_v(s) = (y_{i,j}) \cdot \mathcal{P}_v(0)$, we may rewrite the above as

$$\begin{pmatrix} \iota_v(a) & 0 \\ \iota_v(c) & \iota_v(d) \end{pmatrix} \cdot (y_{i,j}) = \mathcal{P}_v(0) \cdot \begin{pmatrix} \zeta_v & 0 \\ 0 & \xi_v \end{pmatrix} \cdot \mathcal{P}_v(0)^{-1}. \tag{15}$$

Since $\mathcal{P}_v(0) = \text{diag}(\varpi_v, \varpi_v^{-1})$ is diagonal, we easily get

$$\begin{pmatrix} \iota_v(a) & 0 \\ \iota_v(c) & \iota_v(d) \end{pmatrix} \cdot (y_{i,j}) = \begin{pmatrix} \zeta_v & 0 \\ 0 & \xi_v \end{pmatrix}. \tag{16}$$

From this, given that $a \neq 0$ since ϕ_{dR} is invertible, we find $y_{1,2} = 0$. □

Remark 5.2. *Let s be such that $\mathcal{E}_s$ is CM. We note that s will be v-adically close to s_0 with respect to some v that is ordinary for E_0 if and only if the elliptic curves E_0 and $\mathcal{E}_s$ have the same CM-field as their algebra of endomorphisms. Indeed $\text{End}_{\bar{\mathbb{Q}}}(\mathcal{E}_P) \hookrightarrow \text{End}^0_{\bar{\mathbb{F}}_{p(v)}}(\tilde{E}_{0,v})$, for $P \in \{s, s_0\}$, since $\tilde{E}_{0,v} = \tilde{E}_{s,v}$ by our conventions on proximity. On the other hand, since $\mathcal{E}_P$ is CM and $\tilde{E}_{0,v}$ is ordinary these algebras have to be equal for dimension reasons.*

In other words, v-adic relations among the values of our family of G-functions at points of interest for places v that are ordinary for E_0 are only pertinent in the context described in Proposition 5.1.

5.3 Relations at supersingular places

We retain the notation and conventions of Section 5.1 for our family of elliptic curves $f : \mathcal{E} \to S$ and point $s_0 \in S(K)$ whose fiber $E_0 := \mathcal{E}_{s_0}$ is a CM elliptic curve. We let $F_0 := \text{End}^0_{\bar{\mathbb{Q}}}(E_0)$ and consider the set

$$\Sigma_{\text{ssing},unr}(s_0) := \{w \in \Sigma_{K,f} : w \text{ is supersingular for } E_0 \text{ and } v \not| \text{disc}(F_0)\},$$

of supersingular places of E_0 that are over primes unramified in F_0.

Here we restate a theorem of Y. André which is the natural analogue of Proposition 5.1 for places of supersingular reduction of the "center" E_0.

Proposition 5.3 (Y. André [And95], [And03]). *Let $s \in S(\bar{\mathbb{Q}})$ be such that $\mathcal{E}_s$ is also a CM elliptic curve and consider the set of places*

$$\mathcal{P}(s, s_0) := \{v \in \Sigma_{K(s),f} : \exists w \in \Sigma_{\text{ssing},unr}(s_0), v|w \text{ and } s \text{ is } v\text{-adically close to } s_0\}\}.$$

Then there exists a polynomial $R_{s,\text{ssing},unr} \in \bar{\mathbb{Q}}[X_{i,j} : 1 \leq i, j \leq 2]$ and an effectively computable positive constant c_{ssing} depending at most on $f : \mathcal{E} \to S$ and s_0, such that the following hold

1. *$R_{s,\text{ssing},unr}$ has coefficients in the compositum L_s of $K(s)$ with $\text{End}^0_{\bar{\mathbb{Q}}}(\mathcal{E}_s)$,*
2. *$\iota_v(R_{s,\text{ssing},unr}(Y_G(x(s)))) = 0$ for all $v \in \Sigma_{L_s,f}$ for which there exists $w \in \mathcal{P}(s, s_0)$ with $v|w$,*
3. *$R_{s,\text{ssing},unr}$ is homogeneous with $\deg(R_{s,\text{ssing},unr}) \leq c_{\text{ssing}} \cdot |\mathcal{P}(s, s_0)|$, and*
4. *$R_{s,\text{ssing},unr} \notin \langle X_{1,1}X_{2,2} - X_{1,2}X_{2,1} - 1\rangle$.*

Proof. We give a quick sketch of a proof here, especially since the same ideas appear in greater technical detail in the proof of the corresponding fact for QM abelian surfaces in the sequel [Pap25b].

Let d_s be such that $\text{End}^0_{\bar{\mathbb{Q}}}(\mathcal{E}_s) = \mathbb{Q}(\sqrt{-d_s})$. Let us also fix $v \in \Sigma_{L_s,f}$ such that s is v-adically close to s_0 and v is supersingular for E_0 and unramified in $\text{End}^0_{\bar{\mathbb{Q}}}(E_0) =: \mathbb{Q}(\sqrt{-d_0})$. In §5 of [And95] André finds $a_s \in L_s$ (denoted in σ in loc. cit.) and $m_{s,v} \in L_s$ (denoted m_v in loc. cit.) such that the polynomial defined by

$$R_{s,v} := (m_{s,v}+2\sqrt{d_s d_0})X_{1,1}(X_{2,2}+a_s X_{1,2})-(m_{s,v}-2\sqrt{d_s d_0})X_{1,2}(X_{2,1}+a_s X_{1,1}), \tag{17}$$

satisfies $\iota_v(R_{s,v}(Y_G(x(s)))) = 0$.

Consider the matrices $S(n,l) := \begin{pmatrix} n & l \\ 0 & \frac{1}{n} \end{pmatrix} \in \text{SL}_2(\mathbb{C})$ and $T(n) := \begin{pmatrix} 0 & \frac{1}{n} \\ -n & 0 \end{pmatrix} \in \text{SL}_2(\mathbb{C})$. Assume that we had $R_{s,v} \in \langle X_{1,1}X_{2,2} - X_{1,2}X_{2,1} - 1\rangle$. Then $R_{s,v}(S(n,l)) = 0$ implies $nl(4a_s\sqrt{d_0 d_s})+(m_{s,v}+2\sqrt{d_0 d_s}) = 0$ while $R_{s,v}(T(n)) = 0$ gives $m_{s,v} - 2\sqrt{d_0 d_s} = 0$. Since the first of these holds for all n, $l \in \mathbb{C}$ we get $a_s = 0$, $m_{s,v} + 2\sqrt{d_0 d_s} = 0$, and $m_{s,v} - 2\sqrt{d_0 d_s} = 0$. The last two give a contradiction since $d_s d_0 \neq 0$.

The polynomial we want is nothing but $R_{s,\text{ssing},unr} := \prod_v R_{s,v}$, with $R_{s,v}$ as in (17), where the product ranges over all the $v \in \Sigma_{L_s}$ which are such that $|x(s)|_v < \min\{1, R_v(Y_G)\}$ and $v|w$ for some $w \in \Sigma_{\text{ssing},unr}(s_0)$. All properties follow trivially by definition. □

5.4 Relations at archimedean places

In the archimedean case the analogous statement to Proposition 5.3, due to F. Beukers, is the following:

Proposition 5.4 (Theorem 3.3, [Beu93]). *Let $s \in S(\bar{\mathbb{Q}})$ be such that $\mathcal{E}_s$ is also a CM elliptic curve. Then there exists a polynomial $R_{s,\text{arch}} \in \bar{\mathbb{Q}}[X_{i,j} : 1 \leq i, j \leq 2]$ such that the following hold*

1. *$R_{s,\text{arch}}$ has coefficients in the compositum L_s of $K(s)$ with $\text{End}^0_{\bar{\mathbb{Q}}}(\mathcal{E}_s)$,*
2. *$\iota_v(R_{s,\text{arch}}(Y_G(x(s)))) = 0$ for all $v \in \Sigma_{L_s,\infty}$ for which $|x(s)|_v < \min\{1, R_v(Y_G)\}$,*
3. *$R_{s,\text{arch}}$ is homogeneous with $\deg(R_{s,\text{arch}}) \leq c_{\text{arch}} \cdot [\mathbb{Q}(s) : \mathbb{Q}]$, where c_{arch} is some absolute constant, and*
4. *$R_{s,\text{arch}} \notin \langle X_{1,1}X_{2,2} - X_{1,2}X_{2,1} - 1\rangle$.*

Proof. Our reformulation of Beukers' result follows practically identically from the same argument as above. Given $v \in \Sigma_{L_s,\text{arch}}$ with $|x(s)|_v < \min\{1, R_v(Y_G)\}$ Beukers shows that there exist a_s, $m_{s,v}$, $n_{s,v} \in L_s$ such that for the polynomials

$$R_{s,v,1} := X_{1,1}(X_{2,2} + a_s X_{1,2}) - m_{s,v}(X_{1,1}X_{2,2} - X_{1,2}X_{2,1}), \text{ and}$$
$$R_{s,v,2} := X_{1,2}(X_{2,1} + a_s X_{1,1}) - n_{s,v}(X_{1,1}X_{2,2} - X_{1,2}X_{2,1})$$

we have $\iota_v(R_{s,v,j}(Y_G(x(s)))) = 0$.

First note that, since $\det(Y_G(x)) = 1$ generically, if $R_{s,v,1} \in \langle X_{1,1}X_{2,2} - X_{1,2}X_{2,1} - 1\rangle$ we would have, arguing as in the previous proof, $a_s = 0$ and $m_{s,v} = 1$. This may be seen by "testing" at the family $S(n, l)$ for example. But then we get that $X_{1,1}X_{2,2} - 1 \in \langle X_{1,1}X_{2,2} - X_{1,2}X_{2,1} - 1\rangle$ which is trivially false.

The polynomial we want is nothing but the product $\mathbb{R}_{s,\text{arch}} := \prod_v R_{s,v,1}$, where the v range over those archimedean places of L_s for which $|x(s)|_v < \min\{1, R_v(Y_G)\}$. □

5.5 Relations at ramified places

The missing piece in completing the picture in Theorem 1.1 is the construction of relations at points s with $\mathcal{E}_s$ CM elliptic curves among the values of our G-functions at places v that are over primes p ramified in F_0.

Proposition 5.5. *Let $f : \mathcal{E} \to S$ be as in Section 5.1. Let $s \in S(\bar{\mathbb{Q}})$ be such that $\mathcal{E}_s$ is CM.*

Then, there exists a polynomial $R_{s,\mathrm{Ram}} \in \bar{\mathbb{Q}}[X_{i,j} : 1 \leq i,j \leq 2]$ such that the following hold

1. *$R_{s,\mathrm{Ram}}$ has coefficients in some $L_s/K(s)$ with $[L_s : K(s)]$ bounded by an absolute constant,*
2. *$\iota_v(R_{s,\mathrm{Ram}}(Y_G(x(s)))) = 0$ for all $v \in \Sigma_{L_s,f}$ for which $|x(s)|_v < \min\{1, R_v(Y_G)\}$ and $v|\operatorname{disc}(F_0)$,*
3. *$R_{s,\mathrm{Ram}}$ is homogeneous with $\deg(R_{s,\mathrm{Ram}}) \leq c_{\mathrm{Ram}}[K(s) : \mathbb{Q}]$, where c_{Ram} is some absolute constant, and*
4. *$R_{s,\mathrm{Ram}} \notin \langle X_{1,1}X_{2,2} - X_{1,2}X_{2,1} - 1\rangle$.*

Proof. We begin by noting that in any case here we must have that $F_s = F_0$, i.e. $\mathcal{E}_s$ and E_0 have CM by the same CM field. To see this we start by writing $d_P := \operatorname{disc}(\mathrm{End}_{\bar{\mathbb{Q}}}(\mathcal{E}_P))$ for $P \in \{s, s_0\}$. If $d_s = d_{s_0}$ then we are done. If $d_s \neq d_{s_0}$ our assertion follows from Lemma 4.2.

Now we may argue much as in the proof of Proposition 5.1 to find an isogeny $\phi : \mathcal{E}_s \to E_0$ defined over $\bar{\mathbb{Q}}$. The extension L_s will be the compositum of $K(s)$ with the field of definition of this ϕ. The degree $[L_s : K(s)]$ may be bounded by an absolute constant, see for example [Sil92]. From now on we fix $v \in \Sigma_{L_s,f}$ which is such that $v|\operatorname{disc}(F_0)$ and such that our s is v-adically close to s_0.

We write ϕ_{dR} and ϕ_{crys} for the pullbacks of ϕ on the level of de Rham and Crystalline cohomology respectively. Once again we will have that ϕ_{crys} will be the pullback on the level of Crystalline cohomology of the reduction modulo v of ϕ, which we denote by $\tilde{\phi}_v$. As in the proof of Proposition 5.1, by our conventions on v-adic proximity in Definition 3.1, we get that $\tilde{\phi}_v \in \mathrm{End}(\tilde{E}_{0,v})$ since $\tilde{\mathcal{E}}_{s,v} = \tilde{E}_{0,v}$.

Once again we get (13), which we may rewrite, much as in (14), as

$$[\phi]_{dR}\mathcal{P}_v(s) = \mathcal{P}_v(0)[\phi]_{\mathrm{crys}}, \tag{18}$$

where $[\phi]_*$ denotes the matrix induced from ϕ_*, for $* \in \{dR, \mathrm{crys}\}$, with the implied bases on each side of the de Rham-Crystalline comparison isomorphism.

Working much as in the proof of Proposition 5.1, we may assume that $[\phi]_{dR} = \begin{pmatrix} \iota_v(a) & 0 \\ \iota_v(c) & \iota_v(d) \end{pmatrix}$. Setting $y_{i,j} := \iota_v(Y_{i,j}(x(s)))$, since $\mathcal{P}_v(s) = (y_{i,j}) \cdot \mathcal{P}_v(0)$, we may rewrite (18) as

$$\begin{pmatrix} \iota_v(a) & 0 \\ \iota_v(c) & \iota_v(d) \end{pmatrix} \cdot (y_{i,j}) = \mathcal{P}_v(0) \cdot [\phi]_{\text{crys}} \cdot \mathcal{P}_v(0)^{-1}. \tag{19}$$

Taking traces in (19) we conclude that

$$\iota_v(aY_{1,1}(x(s)) + cY_{1,2}(x(s)) + dY_{2,2}(x(s))) = t_v := \text{tr}([\phi]_{\text{crys}}). \tag{20}$$

Practically, all we need at this point is to note that $t_v \in \mathbb{Q}$. This follows from the fact that ϕ_{crys} is the pullback of the endomorphism $\tilde{\phi}_v$ of $\tilde{E}_{0,v}$. See the proof of Lemma 3.5 in [DOP25] for more details.

At this point, much as in the proof of Proposition 5.3, we set $R_{s,v} := (aX_{1,1}+cY_{1,2}+dY_{2,2})^2 - t_v \det(Y_{i,j})$. Note that this way $R_{s,v}$ is a homogeneous degree 2 polynomial and from (20) we have $\iota_v(R_{s,v}(Y_G(x(s)))) = 0$, since $\det(Y_{i,j}(x)) = 1$ on the level of power series. The polynomial we want is nothing but $R_{s,\text{Ram}} := \prod R_{s,v}$, where the product is over all $v \in \Sigma_{L_s,f}$ that divide $\text{disc}(F_0)$ and are such that s is v-adically close to s_0.

To check that $R_{s,\text{Ram}}$ defines a non-trivial relation among the values of Y_G at s we may work again exactly as in Proposition 5.3. In more detail, we need only check that $R_{s,v}$ is not in the ideal $\langle X_{1,1}X_{2,2} - X_{1,2}X_{2,1} - 1 \rangle$. Testing with the same $S(n,l)$ as in Proposition 5.3 is easily seen to force $a = 0$ which contradicts the invertibility of the matrix $[\phi]_{dR}$ corresponding to our isogeny ϕ. □

Remark 5.6. *We note that we need no particular property of the basis of the Crystalline cohomology of the central fiber $H^1_v(E_0)$ in this construction. Rather everything follows from the existence of the isogeny ϕ, which is guaranteed to us by Lemma 4.2.*

6 Height bounds and applications

In this final section we apply André-Bombieri's "Hasse principle for the values of G-functions" to reach the height bounds alluded to in Section 1.2. In the process, we attempt to isolate the difficulty of obtaining the "ideal height bounds" one would want here to conclude an effective version of [Sie14].

6.1 Height bounds

We start by presenting a height bound for the singular moduli of CM elliptic curves whose CM is given by the same CM field as that of a fixed CM elliptic curve. This takes the following form:

Proposition 6.1. *Let j_0 be the j-invariant of a CM-elliptic curve E_0 defined over a number field K and consider the set*

$$A(j_0) := \{j : \mathrm{End}^0_{\bar{\mathbb{Q}}}(E_j) = \mathrm{End}^0_{\bar{\mathbb{Q}}}(E_0)\},$$

of j-invariants whose corresponding elliptic curves, i.e. E_j, have CM by the same CM field as E_0.

Then, there exist effectively computable constants C_1, $C_2 > 0$, depending only on K and j_0, such that

$$h(j) \leq C_1 + C_2 \log([\mathbb{Q}(j) : \mathbb{Q}]),$$

for all $j \in A(j_0)$.

Proof. Without loss of generality, by possibly replacing K by a finite extension K_0/K whose degree depends on j_0, we may assume that E_0 has everywhere good reduction over K, see [ST68].

Let $K(j)$ be the compositum of K and $\mathbb{Q}(j)$. By [Sil92] there exists an extension $K'(j)/K(j)$, with $[K'(j) : K(j)]$ bounded by an absolute constant, such that $\mathrm{End}^0_{\bar{\mathbb{Q}}}(E_j) = \mathrm{End}^0_{K'(j)}(E_j)$ and $\mathrm{End}^0_{\bar{\mathbb{Q}}}(E_0) = \mathrm{End}^0_{K'(j)}(E_0)$. Since $\mathrm{End}^0_{\bar{\mathbb{Q}}}(E_j) = \mathrm{End}^0_{\bar{\mathbb{Q}}}(E_0)$ the two elliptic cures are isogenous over $\bar{\mathbb{Q}}$. By Lemma 6.1 of [MW90] there exists an extension $L(j)/K'(j)$ with $[L(j) : K'(j)] \leq 12$ such that such an isogeny ϕ is defined over $L(j)$.

By Theorem 1.1 of [Paz19] there exist effectively computable positive constants c_1 and c_2, depending at most on j_0, such that

$$h(j) \leq c_1 + c_2 \log(\deg(\phi)). \tag{21}$$

On the other hand, by Theorem 1.4 of [GR14] there exists such an isogeny with $\deg(\phi) \leq \kappa(E_0 \times_K L(j))$ where $\kappa(A)$ is the quantity defined at the beginning of the introduction of loc. cit.. By definition here we have

$$\begin{aligned}\log(\kappa(E_0 \times_K L(j))) \leq c_1' + c_2' \log([L(j) : \mathbb{Q}]) + \\ + c_3' \log(\max\{h_F(E_0 \times_K L(j)), \log([L(j) : \mathbb{Q}]), 1\}).\end{aligned}$$

Now trivially, $\log(\max\{h_F(E_0 \times_K L(j)), \log([L(j) : \mathbb{Q}]), 1\}) \leq \max\{h_F(E_0 \times_K L(j)), \log([L(j) : \mathbb{Q}]), 1\} \leq h_F(E_0 \times_K L(j)) + 1 + \log([L(j) : \mathbb{Q}])$. Since E_0 has, by assumption, everywhere good reduction, and thus also semi-stable reduction, over K, we have by basic properties of the stable Faltings height that $h_F(E_0 \times_K L(j)) = h_F(E_0)$. The conclusion now follows trivially from the above. □

Remark 6.2. *We believe that the above is known to experts. We have opted to include this proof here for two reasons.*

Firstly, the tools used in the proof are either standard in proofs of lower bounds on Galois orbits, or, in the case of Pazuki's result, follow much that same "mantra" as the output of the G-functions method. In other words, the "mantra" of bounding the Weil height of a point of interest by arithmetic data associated to our geometric objects.

The other reason we have opted to include this here, is that it isolates further the problem of obtaining the height bounds envisioned by Y. André in connection to the "Effective Brauer-Siegel" problem.

Before stating our height bound we introduce some notation.

Definition 6.3. *Let $j_0 \neq 0$ be a fixed singular modulus whose corresponding elliptic curve has CM by the CM field F_0. Given another singular modulus j whose corresponding CM elliptic curve has CM given by a CM field $F_j \neq F_0$ we define*

$$\mathcal{P}(j, j_0) := \{v \in \Sigma_{\mathbb{Q}(j,j_0),f} : v(j - j_0) > 0, \tilde{E}_{0,v} \text{ is supersingular}\}. \tag{22}$$

We call $\mathcal{P}(j)$ the set of ***supersingular places of proximity*** *of j to j_0.*

The main output of the G-functions method in this setting is the following height bound:

Theorem 6.4. *Let $j_0 := 54000$ and let $\mathcal{P}(j) := \mathcal{P}(j, 54000)$. Then, there exist effectively computable positive constants c_1 and c_2, such that*

$$h(j) \leq c_1([\mathbb{Q}(j) : \mathbb{Q}] + |\mathcal{P}(j)|)^{c_2}, \tag{23}$$

for all singular moduli j whose corresponding elliptic curve has CM by a field other than $\mathbb{Q}(\sqrt{-3})$.

Proof. We let $f : \mathcal{E} \to S$, where $S := \mathbb{A}^1 \backslash \{0, 1728\}$, be the j-family of elliptic curves, see for example page 47 in [Sil86]. We also fix $j_0 = s_0 = 54000 \in S(\mathbb{Q})$, which is such that[3] $E_0 := \mathcal{E}_{s_0}$ is a CM elliptic curve with CM field $\mathbb{Q}(\sqrt{-3})$. To the pair (f, s_0) we associate the local parameter $x := j - j_0$. To this picture, as discussed in Section 3, we may associate a family of G-functions. In this case, this will be nothing but a single matrix which we denote from now on by Y_G.

Throughout this proof let $j = s \in S(\bar{\mathbb{Q}})$ be another point for which $\mathcal{E}_s$ is CM, with $\mathrm{End}^0_{\bar{\mathbb{Q}}}(\mathcal{E}_s) \neq \mathbb{Q}(\sqrt{-3})$. We consider the set $\Sigma(s) := \{v \in$

[3] See [LMF26, Elliptic Curve 144.1-CMa2].

$\Sigma_{L_s} : |x(s)|_v < \min\{1, R_v(Y_G)\}\}$ of places of the extension L_s considered in Proposition 5.3 with respect to which s is close to s_0. If $\Sigma(s) = \emptyset$ we conclude as in the proof of the main height bound of [Pap22]. From now on we assume $\Sigma(s) \neq \emptyset$ and consider its subsets $\Sigma(s)_{\text{arch}} := \{v \in \Sigma(s) : v|\infty\}$, $\Sigma(s)_{\text{ord}} := \{v \in \Sigma(s) : v \text{ is ordinary for } E_0\}$, and $\Sigma(s)_{\text{ssing}} := \{v \in \Sigma(s) : v \text{ is supersingular for } E_0\}$.

Since $\text{End}^0_{\mathbb{Q}}(\mathcal{E}_s) \neq \mathbb{Q}(\sqrt{-3})$ we can easily see that $\Sigma(s)_{\text{ord}} = \emptyset$, see Remark 5.2. Similarly, from Lemma 4.2 we know that if $v \in \Sigma(s)_{\text{ssing}}$ and p is the unique prime in $\mathbb{Q}$ which is divisible by v, then p is unramified in $\mathbb{Q}(\sqrt{-3})$. In other words, $p \neq 3$ the unique prime ramified in $\mathbb{Q}(\sqrt{-3})$. In particular, we may use Proposition 5.3 to get a polynomial $R_{s,\text{ssing}} \in L_s[X_{i,j} : 1 \leq i, j \leq 2]$ such that $\iota_v(R_{s,\text{ssing}}(Y_G(x(s)))) = 0$ for all $v \in \Sigma(s)_{\text{ssing}}$.

The places of the set $\Sigma(s)_{\text{arch}}$ are "dealt with" by the polynomial $R_{s,\text{arch}}$ of Proposition 5.4. The polynomial $R_s := R_{s,\text{arch}} \cdot R_{s,\text{ssing}}$ will satisfy

$$\iota_v(R_s(Y_G(x(s)))) = 0$$

for all places $v \in \Sigma(s)$, thanks to our comments above. This polynomial will correspond to a global non-trivial relation among the values of the G-functions of our family at the point s. We may now apply André-Bombieri's "Hasse principle for values of G-functions", see Chapter $VII.5$ in [And89]. The result now follows since $\deg(R_s)$ is bounded by the quantity on the right of (23). □

We note that the choice of j_0 here is completely arbitrary, other than the condition $j_0 \neq 0, 1728$, as well as the choice of the CM field $\mathbb{Q}(\sqrt{-3})$. A different choice of j_0 would alter the constant c_1 in (23), since c_1 depends on arithmetic quantities of the fiber $\mathcal{E}_{s_0}$.

The same exact argument as above would give us the slightly more general:

Theorem 6.5. *Let $f : \mathcal{E} \to S$ be a 1-parameter family of elliptic curves defined over a number field K and $s_0 \in S(K)$ a point whose fiber $\mathcal{E}_{s_0}$ has CM by the CM field F_0. Assume that the induced map $S \to Y(1)$ is not-isotrivial.*

Then there exist effectively computable positive constants C_1, C_2, depending on $f : \mathcal{E} \to S$ and s_0, such that

$$h(s) \leq C_1([K(s) : K] + |\mathcal{P}(j(s), j(s_0))|)^{C_2}, \tag{24}$$

for all $s \in S(\bar{\mathbb{Q}})$ whose fibers have CM given by some CM field other than F_0, and where $j(P)$ denotes the j-invariant of the fiber $\mathcal{E}_P$.

6.2 The effective Brauer-Siegel problem

One of André's motivations towards his seminal work in [And89], seems to have been, see in particular Remark 3 on page 201 of loc. cit., the search for an effective version of the following classical, and famously ineffective, result of C. L. Siegel:

Theorem 6.6 (Siegel,[Sie35]). *Given $\epsilon > 0$ there exists a constant $c(\epsilon) > 0$ such that*

$$h(K) \geq c(\epsilon)|\operatorname{disc}(K)|^{\frac{1}{2}-\epsilon}, \tag{25}$$

for all imaginary quadratic number fields K.

Via standard results about CM elliptic curves, Theorem 6.6 may be translated to a statement about CM elliptic curves. Given a singular modulus j we let E_j be the corresponding CM elliptic curve and set $D(j) := \operatorname{disc}(\operatorname{End}_{\bar{\mathbb{Q}}}(E_j))$ to be the discriminant of the ring of $\mathbb{Q}$-endomorphisms of E_j. An effective answer to Theorem 6.6 would be equivalent[4] to finding an effective constant $c(\epsilon) > 0$ such that

$$[\mathbb{Q}(j) : \mathbb{Q}] \geq c(\epsilon)|D(j)|^{\frac{1}{2}-\epsilon}. \tag{26}$$

The proof of Theorem 1.5 in [Pap26], replacing the usage of the height bounds in Theorem 2.15 there by the height bound in Theorem 6.4, leads to the existence of effectively computable constants c_4, c_5 for which

$$|D(j)| \leq c_4 \max\{[\mathbb{Q}(j) : \mathbb{Q}], [\mathbb{Q}(j) : \mathbb{Q}] + |\mathcal{P}(j)|\}^{c_5}, \tag{27}$$

for all singular moduli j for which $\operatorname{End}^0_{\bar{\mathbb{Q}}}(E_j) \neq \mathbb{Q}(\sqrt{-3})$. Compare, in particular, equation (52) in loc. cit. with (27). Furthermore note here that, were we to use Proposition 6.1 instead of Theorem 6.4, we would get effective constants dealing with the j with $\operatorname{End}^0_{\bar{\mathbb{Q}}}(E_j) = \mathbb{Q}(\sqrt{-3})$, albeit with a fractional power on the right of (26) slightly worse than $\frac{1}{2}$.

The obvious question that now arises is if the contribution of $|\mathcal{P}(j)|$ can be made "negligible". What one would want here is to find effectively computable constants C_1, C_2, and $C_3 > 0$ such that, for all singular moduli j with $\operatorname{End}^0_{\bar{\mathbb{Q}}}(E_j) \neq \mathbb{Q}(\sqrt{-3})$ we have

$$|\mathcal{P}(j)| \leq C_1[\mathbb{Q}(j) : \mathbb{Q}]^{C_2} \cdot D(j)^{\frac{1}{C_3}}, \tag{28}$$

with $C_3 > c_5$.

We close off with some remarks on the constant c_5 of (27). First of all, in this case, the constant c_5 will be the product of the constant c_1 outputted by

[4] We point the interested reader to the introduction of [Pap26] for more on this.

André-Bombieri's "Hasse-Principle" and a constant c_1' outputted by Masser-Wüstholz's "Endomorphism estimates", see [MW94].

In [MW94] Masser and Wüstholz note that the optimal value of the constant c_1' is of the magnitude $2+\epsilon$ in the case of elliptic curves. André's method on the other hand would give $c_1 = 9$. We note that, following Remark 2 on page 138 of [And89], we may even take $c_1 = 4$, i.e. using the height bound for "strongly non-trivial relations" in the vocabulary of André's version of the "Hasse principle" in loc. cit.. All in all, it would therefore suffice for our purposes to have a version of (28) with $C_3 > 8$.

Remark 6.7. *The above, perhaps naive, sketch of a strategy towards an effective version of Siegel's result, is at first glance at least, crude from the perspective of the "G-functions method". More precisely, the problematic quantity in the height bounds in Theorem 6.4 is there to control the way that the relations outputted by Proposition 5.3 depend on the place in question. A, perhaps, more reasonable approach would be to try to directly bound the number of the possible polynomials that might appear in Proposition 5.3, by, say, the same quantity as in* (28).

This approach was, in principle at least, considered by Beukers, see in particular Theorem 5.4 *in [Beu93]. Using Beukers' results, the height bound in Theorem 6.4 would look like*

$$h(j) \leq c_0[\mathbb{Q}(j) : \mathbb{Q}]^4 \cdot D(j)^2. \tag{29}$$

Were we to pair this with Masser-Wüstholz's estimates, we would reach a clear impasse.

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